



01 — Business Context and Growth Strategy

TASCO Insurance × VETC Growth Platform — Business & Product

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How to read this document. Statements about TASCO, VETC and the market come **only** from the client brief. Statements about the solution are grounded in the codebase; file paths and API endpoints are cited. Everything else is an **assumption**, labelled [A-*nn*] and listed in §13. All financial figures are **illustrative** and must be replaced with TASCO data during discovery.

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1. Executive summary

The situation. VETC, Tasco group's electronic toll collection operator, has about 6 million car drivers, and 70–80% of them use the VETC app. That is one of the largest direct-to-driver relationships in Vietnam. Today VETC sells compulsory motor third-party liability insurance (*Bảo hiểm TNDS bắt buộc*) mainly through agents, banks, car showrooms and corporate partners. Its own channels (app push, telesales) are small. **Zero renewals have been closed through own channels this year.**

The diagnosis. The problem is not demand. TNDS is compulsory, and drivers need a valid certificate for inspection (*đăng kiểm*) and roadside checks. The near-zero result has three root causes:

1. **Data.** Only 1 in 10 records has a valid policy stamp, so VETC does not know *when* to contact most drivers.
2. **Trust.** Customers do not trust sales calls in a market full of scam calls.
3. **Value.** TNDS premiums are fixed by regulation and direct discounts are illegal, but most customers are looking for a percentage discount. Partners close faster, and the app cannot renew yet.

The answer. We propose a **growth platform** that wins on *data, trust and service*, never on price. It covers **new business as well as retention**:

Lever	What the platform does	Evidence in code
Data repair loop	Merges dirty multi-source records into one golden profile per plate. It infers policy expiry with a stated confidence, and every customer touch (app confirmation, bot call, purchase) improves the record.	<code>src/domain/enrichment.js</code> , <code>config/rules/enrichment.json</code> , <code>POST /api/customer/expiry</code>
Explainable prioritisation	Scores every vehicle 0–100 with plain-language reasons and picks a next-best action and journey.	<code>src/domain/leads.js</code> , <code>config/rules/scoring.json</code> , <code>nba.json</code> , <code>journeys.json</code>
Trust-first contact	The voice bot discloses that it is automated, verifies the plate first (the customer says it; the bot never reads it out) and never takes payment. Payment happens only in the VETC app. Certificates can be checked by QR.	<code>src/domain/voicebot.js</code> , <code>config/rules/content.voicebot.json</code> , <code>GET</code> <code>/api/public/certificates/:certNo</code>
One-tap renewal	Signed deep link, then quote, then VETC wallet debit, then TASCO e-certificate. Idempotent, so a customer can never be charged twice.	<code>src/application/salesService.js</code> , <code>POST /api/customer/quotes</code> , <code>POST</code> <code>/api/customer/orders</code>
Value beyond discount	Roadside assistance, e-certificate, inspection reminders, claims fast-lane, multi-year cover and cover upgrades. A copy guard blocks the word "discount" from customer copy.	<code>config/rules/benefits.json</code> , <code>copy_guard.json</code>
Partners as a channel	A partner API with plate-based quoting, binding and commission statements, capped at statutory limits.	<code>/api/partner/v1/*</code> , <code>config/rules/commission.json</code>

Illustrative base case (§12; every input is an assumption). At full scale in year 1 after the MVP, own channels plus the partner API issue about **140,000 TNDS policies**, worth about **VND 79 bn net premium (≈ USD 3.1 M)**. Cross-sell adds about **VND 6.9 bn**. The AI voice bot does the qualifying work for about **VND 1.2 bn**; the same work done by human telesales would cost about **VND 13.4 bn**. The MVP can be live with a pilot cohort in **8–10 weeks** (§10).

Our recommendation: approve an MVP pilot in 1–2 provinces on a 50,000–100,000-vehicle cohort. Decide on scale-up with measured conversion, data-repair and trust KPIs (§11).

2. Market and TASCO/VETC context

2.1 The players (from the brief)

Entity	Role	Relevance
TASCO Insurance (tascoinsurance.com.vn)	Non-life insurer in Vietnam. Subsidiary of Tasco Joint Stock Company, which acquired Groupama Vietnam General Insurance in 2022.	Risk carrier and issuer of TNDS, physical damage and personal accident policies. Owns underwriting, tariffs and claims.
Tasco JSC	Group parent.	Owns both TASCO Insurance and VETC, so the group already has a captive distribution opportunity.
VETC (vetc.com.vn)	Electronic toll collection operator with an e-wallet-like platform. About 6 M car drivers; 70–80% use the VETC app.	Owns the customer relationship, the app, the wallet, toll and trip signals, and tag activation events.
Other insurers on VETC	VETC is a multi-insurer platform (TASCO, PVI, PTI via ADD Solutions). Customers can choose.	TASCO must <i>earn</i> the choice through service. VETC's neutrality must be respected (§16).

Entity	Role	Relevance
Traditional channels	Agents and partners (individuals and companies), banks, car showrooms, corporate and fleet accounts. Partners close most deals today.	The current engine of volume. They should be armed, not displaced (\$5).

2.2 Product and regulatory frame

- **TNDS is compulsory and price-regulated.** The platform's tariff rules carry the annual premiums from Decree 67/2023/ND-CP Appendix I (`config/rules/tariff.tnds_car.json` , `tariff.tnds_motorbike.json`), for example VND 437,000 + 10% VAT = **VND 480,700** for a car under 6 seats. *TASCO underwriting must confirm the figures before go-live.*
- **Discounts and rebates are prohibited** in customer copy (Law on Insurance Business 08/2022/QH15, cited in `config/rules/copy_guard.json`). The platform therefore competes on **service and convenience**.
- **Partner commission has statutory caps.** These are encoded per product in `config/rules/commission.json` (for example TNDS 5% of net premium). *TASCO finance and legal must confirm.*
- **Contact and privacy obligations:** anti-spam rules (Decree 91/2020/ND-CP), Decree 13/2023/ND-CP on personal data protection, and the Personal Data Protection Law 91/2025/QH15 (effective 2026). These are encoded conservatively in `config/rules/contact_policy.json` and `retention.json` . *Confirm with TASCO legal.*

2.3 Strategic asset inventory

Asset	Owner	How the platform uses it
6 M driver relationships, app reach 70–80%	VETC	Primary reach: app push costs VND 0 per send (<code>config/rules/costs.json</code>)
ETC tag activation events	VETC	Signal of a new vehicle, which triggers the <i>new_vehicle</i> journey (<code>config/rules/triggers.json</code>)
Toll trips, long trips, app sessions	VETC	Engagement score and benefit relevance (<code>scoring.json</code> , <code>benefits.json</code>)
VETC wallet and auto top-up	VETC	One-tap payment; "wallet covers premium" affinity signal
Inspection booking	VETC (service)	Moment of truth: valid TNDS is required at inspection
Policy administration, e-certificates, claims	TASCO	Issuance via the <code>policyAdmin</code> port; QR verification; FNOL
Partner network	VETC / TASCO	Partner API and portal, commission statements

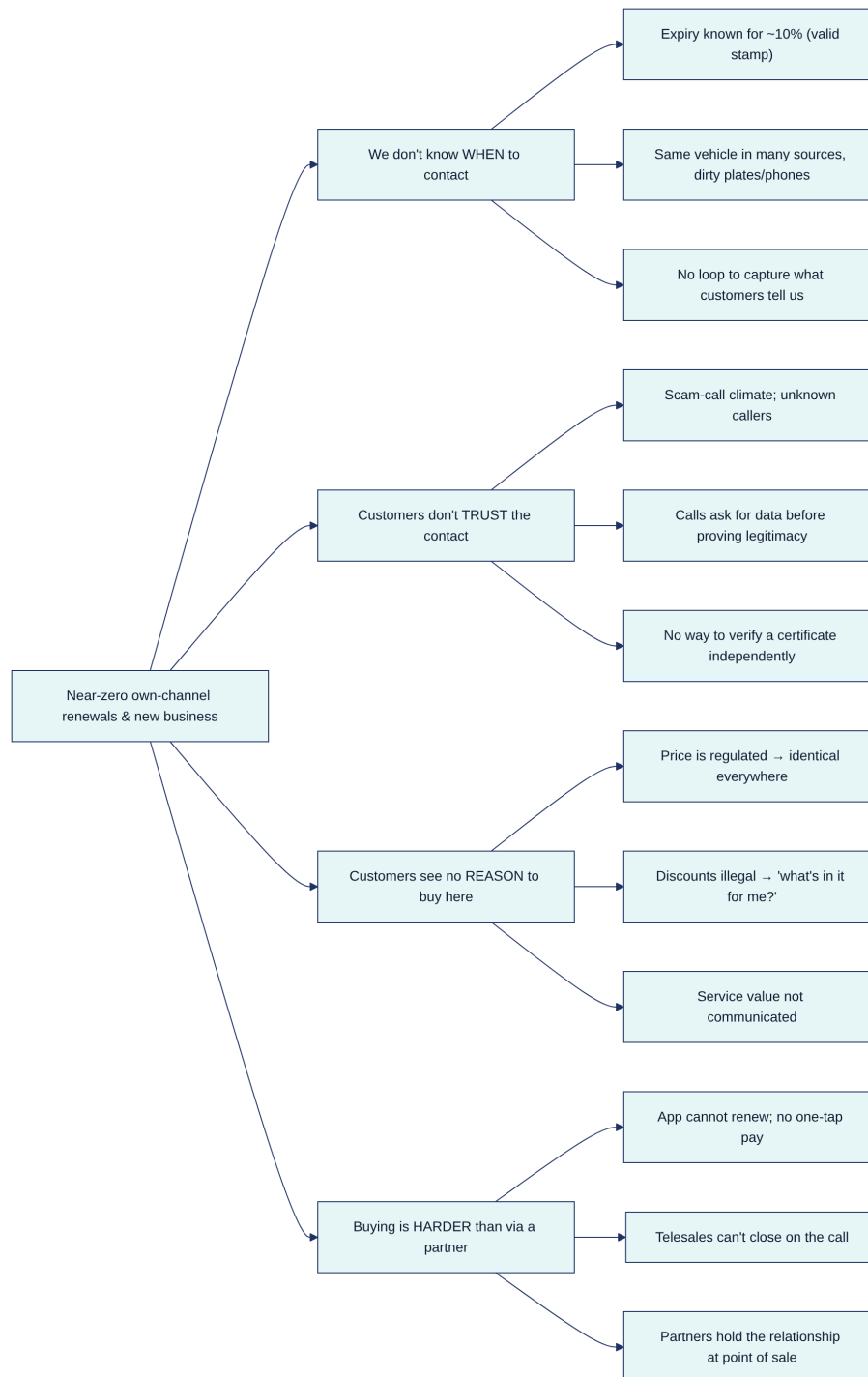
3. Problem analysis: why own-channel renewals are near zero

3.1 Symptoms (from the brief)

- Only **1 in 10** records has a valid policy stamp.
- **0** renewals were closed through own channels this year.
- Telesales converts poorly. Customers distrust sales calls, direct discounts are illegal, partners close faster, and the app cannot renew.

- Most customers care about a percentage discount.

3.2 Root-cause tree



3.3 Root causes and design responses

#	Root cause	Design response	Where it lives
R C- 1	Expiry date unknown or unreliable for most vehicles	Evidence-based expiry inference with confidence (verified certificate 1.0, customer-declared 0.75, partner record ≤ 0.7 , inspection cycle 0.5, tag anniversary 0.25) plus corroboration boosts. Low-confidence vehicles get a <i>verify expiry</i> action before any selling.	<code>enrichment.json</code> → <code>expiryEvidence</code> , <code>nba.json</code> rule <code>fix_data</code>

#	Root cause	Design response	Where it lives
R C- 2	Duplicate, dirty multi-source records	Plate and phone normalisation; one golden record per plate; source-trust survivorship; field-level lineage	<code>src/domain/identity.js</code> , <code>enrichment.js</code> , <code>GET /api/customers/:id/lineage</code>
R C- 3	No capture loop	Customer confirms expiry in the app, the bot captures "already renewed", insurer and next date, and TASCO issuance sets confidence to 1.0	<code>POST /api/customer/expiry</code> , <code>voiceService.finalize</code> , <code>salesService.purchase</code>
R C- 4	Distrust of calls	Bot disclosure, plate-first verification, a "never OTP or payment" statement, a trust script for "is this a scam?", payment only in the app	<code>content.voicebot.json</code> lines <code>intro</code> , <code>trust</code> , <code>plateMismatch</code>
R C- 5	No independent proof	Public certificate verification page and QR with no PII	<code>GET /api/public/certificates/:certNo/verify/:certNo</code>
R C- 6	Price-only mindset	Benefits engine ranks the value items most relevant to each driver; the voice bot answers price truthfully ("regulated, same everywhere") and pivots to value	<code>benefits.json</code> , <code>content.voicebot.json</code> line <code>price</code>
R C- 7	Hard to buy	Signed deep link → quote → VETC wallet debit → e-certificate; idempotent	<code>POST /api/customer/session</code> , <code>/quotes</code> , <code>/orders</code>
R C- 8	Partners own the point of sale	Partner API to quote and bind by plate, with transparent commission	<code>/api/partner/v1/*</code>

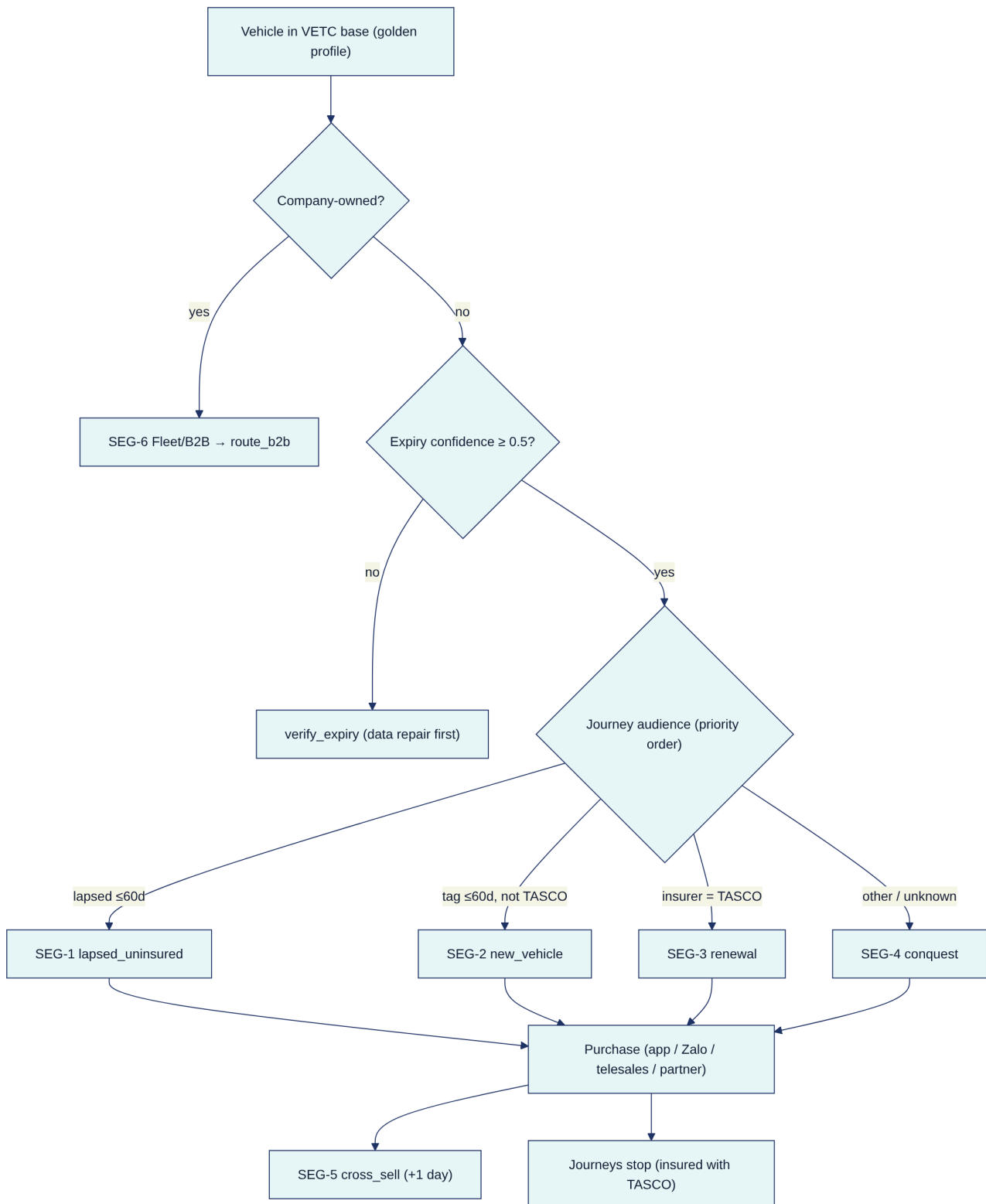
4. Growth strategy: new business and retention

The client asked us not to limit scope to TNDs renewals. The strategy therefore treats the VETC base as a **portfolio of segments**, each with its own journey, objective and economics. The journey engine places each vehicle in the **first matching journey by priority** (`config/rules/journeys.json`). It stops all contact as soon as the vehicle is insured with TASCO (`leadService.recomputeOne` → `action.insured`).

4.1 Segment portfolio

Seg ID	Segment	Definition (rule)	Objective	Journey ID	Primary play
SEG -1	Uninsured / lapsed vehicles	Expiry passed within the last 60 days with confidence ≥ 0.5	New business	<code>lapsed_uninsured</code> (priority 1)	Service notice ("may be driving without compulsory TNDs"), then a bot call (hot/warm), then telesales (hot)
SEG -2	New vehicles	ETC tag activated ≤ 60 days ago, insurer not TASCO	New business	<code>new_vehicle</code> (priority 2)	Welcome, save TNDs in the app, verify expiry, value reminder

Seg ID	Segment	Definition (rule)	Objective	Journey ID	Primary play
SEG -3	TASCO book renewal	Current insurer = TASCO	Retention	renewal (priority 3)	Seven-step cadence from -45 to 0 days: verify, remind, value, bot, urgent, telesales, expiry day
SEG -4	Conquest	Insured elsewhere or insurer unknown	New business	conquest (priority 4)	Same cadence, using the conquest_reminder template and benefit-led copy; bot for hot leads only
SEG -5	Cross-sell and bundles	Bought TNDs only and gave marketing consent	Cross-sell	cross_sell (post-purchase, +1 day)	PA_SEAT (passengers), MOTOR_PD (own damage), bundles SAFE_DRIVE and FULL_MOTOR
SEG -6	Fleet / B2B	ownerType = company	New business and retention	NBA route_b2b	Route to the B2B team; fleet dashboard and consolidated invoice (benefit fleet_dashboard)
SEG -7	Partner-led	Sold by a bank, showroom, agent, fleet operator or inspection centre	New business	Partner API	Partners quote and bind by plate; VETC data pre-fills; commission statement



Note: the decision order above is simplified from `nba.json`, where `dnc` → `fleet` → `fix_data` → ... is evaluated first-hit. Journey assignment (`journeys.json`) runs independently and in priority order. See §16 risk R-08 on fleet vehicles entering B2C journeys.

4.2 Segment play summary

Segment	Why they buy here	Hook (copy key)	Best channel order	Escalation
SEG-1 Lapsed	Legal exposure, and inspection blocked without TNDS	<code>lapsed_notice</code> (service, not marketing)	Push → Zalo ZNS → SMS	Bot +2 days (hot/warm), telesales +5 days (hot)
SEG-2 New vehicle	First time in the ecosystem; set-and-forget reminders	<code>new_vehicle_welcome</code> , <code>verify_expiry</code>	Push → Zalo	Value reminder +14 days
SEG-3 Renewal	Already a TASCO customer; one-tap renewal	<code>first_reminder</code> , <code>urgent_reminder</code> , <code>expiry_day</code>	Push → Zalo → SMS	Bot -14 days (hot/warm), telesales -3 days (hot)
SEG-4 Conquest	Same regulated price, <i>better service</i>	<code>conquest_reminder</code> , <code>value_reminder</code>	Push → Zalo → SMS	Bot -14 days (hot only)
SEG-5 Cross-sell	"TNDS only covers third parties"	<code>cross_sell</code>	Push → Zalo	None (marketing consent required)
SEG-6 Fleet	One view and one VAT invoice	Account management	B2B team	—
SEG-7 Partner	Partner closes faster with VETC data	API	Partner's own front end	Commission

4.3 Event-driven moments of truth

Calendar-driven reminders are reinforced by **ecosystem events**, the moments when a driver is most receptive (`config/rules/triggers.json`, `POST /api/ecosystem/events`):

Trigger ID	VETC event	Condition	Action	Marketing?
<code>tag_activated</code>	<code>vetc.tag_activated</code>	—	Enrol in <code>new_vehicle</code>	—
<code>inspection_blocked</code>	<code>vetc.inspection_blocked</code>	Expiry < 60 days away	<code>inspection_tnds_check</code> via push or Zalo	No (service)
<code>wallet_topup</code>	<code>vetc.wallet_topped_up</code>	Expiry within ±30 days	<code>first_reminder</code> push: customer is in the app with funds	Yes
<code>long_trip</code>	<code>vetc.long_trip_started</code>	Lapsed ≤ 60 days	<code>value_reminder</code> push	Yes

5. Channel strategy: arm the partners, don't fight them

5.1 Principle

Partners close most TNDS deals today because they are present at the moment of need: a car purchase, a loan, an inspection. Competing with them on the same moment is slow and damages the relationship. The strategy is to **own the moments partners do not see** (renewal timing, in-app moments, lapsed vehicles) and to **arm partners** with VETC data and instant TASCO issuance for the moments they do own.

5.2 Channel roles

Channel	Code (channel)	Role	Cost per contact (costs.json)	Consent needed (contact_policy.json)
VETC app push	app_push	Primary reminder; one-tap renew	VND 0	marketing for marketing messages
Zalo ZNS / OA	zalo_zns	Secondary reminder; trusted branded channel	VND 300	marketing for marketing messages
SMS	sms	Fallback for phone-only contacts	VND 700	marketing for marketing messages
AI voice bot	voice_bot	Qualify, verify the plate, capture data, hand off or send a link	VND 1,500/min × 1.6 min = VND 2,400 per call	call (+ marketing)
Telesales	telesales	Close hot leads; handle objections	VND 6,000/min × 4.5 min = VND 27,000 per call	call (+ marketing)
Partner API	partner_api	Partner-originated new business	Commission (TNDS 5% of net)	Partner obtains consent (consentMarketing field)

Service messages (lapsed notice, expiry day, purchase confirmation, a link the customer asked for) bypass frequency caps but still respect DNC and channel availability (`serviceMessagesBypassCaps: true`). Marketing messages are limited to **08:00–20:00 ICT, 1 per day and 3 per week**, with at most **2 call attempts per week** (`contact_policy.json`).

5.3 Waterfall logic

For each touchpoint the journey engine tries the step's channels in order. It sends on the **first permitted channel that succeeds** and records the reasons for any channel it blocks (`journeyService.executeTouchpoint`). Low-cost digital channels therefore always go first, and calls are reserved for hot and warm leads (`onlyTiers`).

5.4 Partner enablement

Capability	Endpoint	Business value
Onboard partner by type (bank, showroom, agent, fleet, inspection centre)	POST /api/partners	Self-serve partner growth
Issue or revoke API key (shown once, stored hashed)	POST /api/partners/:id/keys , DELETE /api/partners/keys/:keyId	Secure B2B integration
Quote by plate; unknown vehicles are onboarded as new business	POST /api/partner/v1/quotes	Each partner sale also enriches the VETC base
Bind (idempotent)	POST /api/partner/v1/orders	Instant e-certificate at the counter
Own policies and commission statement	GET /api/partner/v1/policies , /statement	Transparent and dispute-free commission
Commission by product × partner type, capped at statutory caps	config/rules/commission.json	Compliance by construction

6. Value beyond discount

Price cannot move, so the proposition is built from **services, convenience and relevant cover**. The benefits engine (`config/rules/benefits.json` , `src/domain/leads.js#benefitsFor`) ranks items by relevance to each driver, explains *why* in plain language, and shows customers at most 3 items. Only items with `legalStatus = approved` reach customers. Staff also see pending items, flagged as pending.

Benefit ID	Type	Proposition (EN / VI)	Personalisation signal	Legal status
<code>roadside_24_7</code>	Service	24/7 roadside assistance / <i>Cứu hộ giao thông 24/7</i>	Long trips (km over 90 days), toll trips	Approved
<code>e_certificate</code>	Service	Instant e-certificate + QR check / <i>Giấy chứng nhận điện tử + mã QR</i>	Universal	Approved
<code>auto_renew</code>	Service	Never-lapse auto renewal (confirm-before-debit)	Wallet auto top-up	Approved (<i>feature planned; see doc 02 FR-061</i>)
<code>inspection_assist</code>	Service	Inspection reminder and booking / <i>Nhắc lịch và đặt lịch đăng kiểm</i>	Expiry inferred from inspection cycle	Approved
<code>claims_fast_lane</code>	Service	In-app accident reporting / <i>Báo tai nạn nhanh</i>	Toll trips	Approved
<code>loyalty_points</code>	Loyalty	Non-cash VETC points	App sessions	Pending legal review
<code>multi_year</code>	Convenience	2–3-year cover at regulated pro-rata premium	Wallet balance > VND 1 M; individuals only	Approved
<code>upsell_pass_seat</code>	Cover upgrade	Protect your passengers (PA per seat)	Seats ≥ 7	Approved
<code>upsell_motor_pd</code>	Cover upgrade	Cover damage to your own car	Vehicle age ≤ 5	Approved
<code>fleet_dashboard</code>	Service	Fleet renewal dashboard and consolidated VAT invoice	Company owner	Approved (<i>feature planned</i>)

Guardrails.

- The **copy guard** (`copy_guard.json`) bans "giảm giá", "chiết khấu", "hoàn tiền", "khuyến mãi phí", "rẻ hơn", "discount", "cashback", "rebate", "% off", "cheaper", "lower premium" and "price cut". It is enforced twice: when a content rule set is saved (`src/rules/validators.js`) and again at send time (`journeyService.sendMessage` → message status `blocked`).
- Referral** (`referral.json`) is non-cash VETC points only and is **disabled** until legal approval.
- Bundles** (`SAFE_DRIVE = TNDS + PA_SEAT`; `FULL_MOTOR = TNDS + MOTOR_PD + PA_SEAT`) are priced as the **sum of components** (`products.json`). A bundle is a convenience, not a discount.

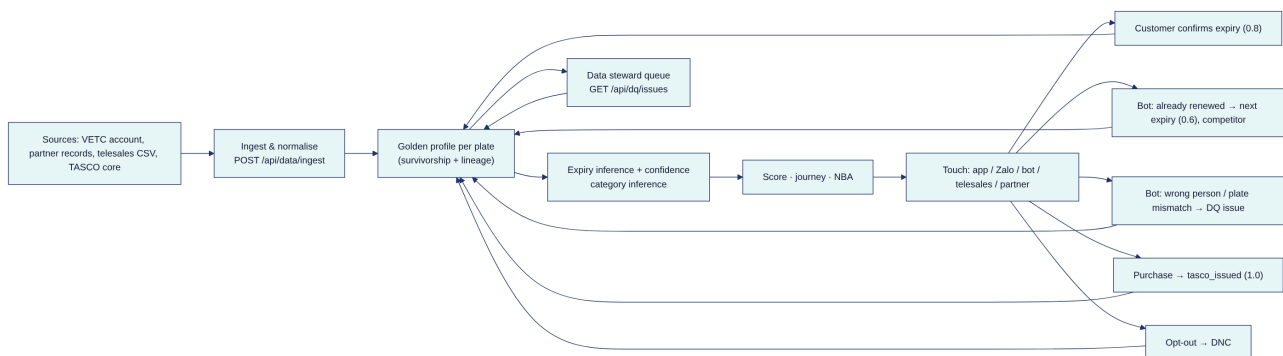
7. Trust strategy

Trust principle	Mechanism	Code / config
Prove legitimacy before asking anything	The bot says it is VETC's <i>automated assistant</i> , states the call is recorded, and says "VETC never asks for OTP or payment by phone"	<code>content.voicebot.json</code> → <code>disclosure</code> , line <code>intro</code>

Trust principle	Mechanism	Code / config
Plate-first verification	The customer says the plate; the bot extracts it from speech and compares it with the record. It never reads the plate out. Mismatch → the call ends politely. 3 attempts maximum.	<code>src/domain/identity.js#extractPlateFromSpeech</code> , <code>voicebot.js state verify_plate</code> , <code>maxPlateAttempts: 3</code>
Masked data on calls	Plate masked (30A-***.45) and phone masked in summaries	<code>voicebot.js#maskPlate</code> , <code>handoffSummary</code>
No payment on calls	Bot and telesales send a link; payment happens only in the VETC app or Zalo OA via wallet	Line <code>link</code> ; talking points in <code>handoffSummary</code>
Scam objection handled, not ignored	Intent <code>scam_concern</code> → <code>trust</code> script and an offer to send an in-app notice the customer can check	<code>content.voicebot.json</code> intents and lines
Independently verifiable certificate	QR on the e-certificate opens a public verification page showing the masked plate, validity and insurer, with no PII	GET <code>/api/public/certificates/:certNo</code> , <code>/verify/:certNo</code>
Branded channels only	Messages come from the VETC app and VETC's official Zalo OA. Links are HMAC-signed (no enumerable IDs).	<code>bootstrap/container.js</code> → <code>links.sign/verify</code>
Respect "no"	Opt-out on a call → <code>dnc = true</code> , call consent withdrawn, audit entry. Consent centre in the app.	<code>voiceService.finalize case opted_out</code> , PUT <code>/api/customer/consent</code>
Honest price talk	"TND\$ premiums are set by regulation and are the same at every insurer"	Line <code>price</code> ; quote note in <code>rating.js</code>

8. Data strategy: every touch improves the data

8.1 The data-repair loop



8.2 Data-quality model

- DQ issue types:** `phone`, `name`, `reliable_expiry`, `vehicle_category`, `current_insurer` (missing), `conflicting_phone`, `invalid_plate`, `wrong_person`, `plate_mismatch` (`enrichment.js`, `ingestionService.js`, `voiceService.js`).
- DQ score per profile** = $100 \times (0.5 \times \text{completeness} + 0.35 \times \text{expiry confidence} + 0.15 \times \text{category confidence})$ (`enrichment.json` → `dataQuality`).
- Usable expiry threshold:** confidence ≥ 0.5 (`usableExpiryConfidence`). Below this threshold the next-best action is *verify expiry*, not *sell*.

- **Platform-captured facts survive source rebuilds.** Customer-declared, bot-captured and TASCO-issued expiries, and consent overrides, are not overwritten by lower-confidence batch data (`ingestionService.rebuild`).

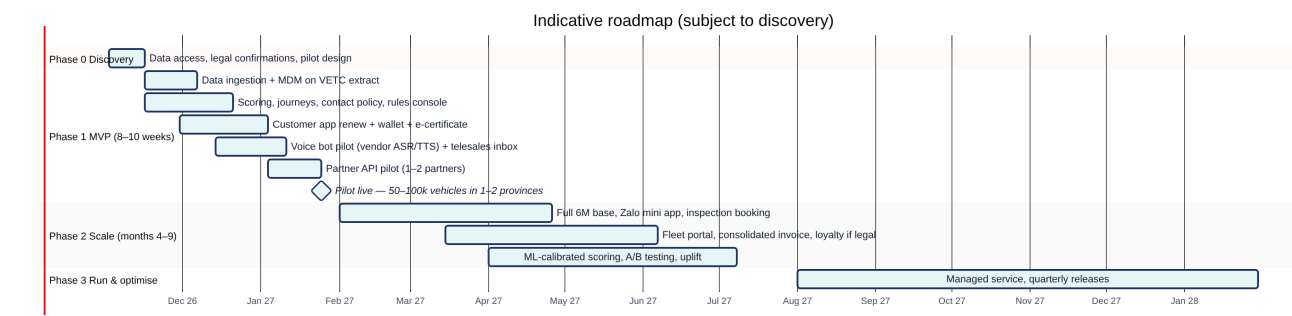
8.3 Data governance principles

1. **Lineage for every field.** Each field records its source, confidence and, where applicable, the rule that set it (`GET /api/customers/:id/lineage`).
2. **Minimum necessary data.** Telesales handoffs carry only what is needed to call; PII is masked unless the role holds `profile:read_pii`.
3. **Encrypted PII.** Name, phone, transcripts and claim descriptions are encrypted at rest (AES-256-GCM). Phone search uses an HMAC blind index (`src/adapters/persistence/schema.js`).
4. **Retention by rule.** Retention periods are set per entity in `config/rules/retention.json`.

9. How the strategy maps to the four challenge tracks

Track	Brief ask	Platform capability	Beyond the brief	Main FR IDs (doc 02)
T1 AI voice bot	Call interested and expiring leads, confirm the plate first, hand hot leads to telesales	Dialogue engine with disclosure, plate-first verification, expiry confirmation, 12 intents, price-honest script, 9 outcomes, structured handoff, campaign runner, governance KPIs	Outcomes write back to data (already renewed → next expiry; opt-out → DNC; mismatch → DQ issue). Scripts are versioned, copy-guarded rules.	FR-031 – FR-039
T2 Lead scoring and enrichment	Build profiles from incomplete records; rank who to call first	MDM golden record, evidence-based expiry inference, category inference, explainable 5-factor score, next-best action	Lineage, DQ queue, steward corrections, simulation before rule changes	FR-001 – FR-019, FR-085
T3 Renewal engine	Automatic reminders; simple purchase via VETC app and Zalo	Four journeys and cross-sell, contact policy, copy guard, signed links, one-tap wallet purchase, e-certificate	New-business journeys (lapsed, new vehicle, conquest), ecosystem triggers, partner API	FR-020 – FR-030, FR-045 – FR-052, FR-074 – FR-077
T4 Value beyond discount	Roadside, loyalty, bundles instead of price cuts	Benefits catalogue with legal gating, relevance and "why", bundles, multi-year cover, cover upgrades, claims FNOL	Copy guard enforcement; QR verification; fleet proposition	FR-053 – FR-062, FR-071 – FR-073

10. Phased roadmap

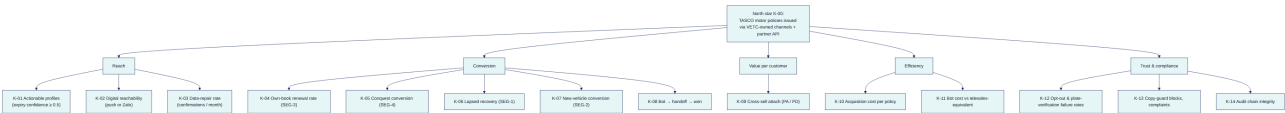


Phase	Duration	Scope	Exit criteria (proposed)
0 Discov ery	2 weeks	Data extract and schema mapping (VETC accounts, partner lists, TASCO core), legal confirmations (tariff, commission caps, contact policy, PDP), integration contracts (wallet, policy admin, Zalo ZNS, SMS brandname, voice vendor), pilot cohort design with control group	Signed pilot design; data-sharing agreement; integration sandbox access
1 MVP	8–10 weeks	Everything in the current codebase moved from sandbox ports to real adapters for one voice vendor, VETC wallet, TASCO core and Zalo ZNS. Staff console and customer app. Pilot cohort.	Pilot live; KPIs K-01 – K-12 instrumented; security test passed; UAT sign-off
2 Scale	~6 months	Full 6 M base with incremental ingestion; Zalo mini app; inspection booking integration; fleet dashboard; auto-renew opt-in; loyalty (after legal approval); ML score calibrated against outcomes; experimentation framework	Base-case KPI targets on track; cost per policy ≤ target (K-10)
3 Run	Ongoing	Managed service, rule tuning by the business through the maker-checker console, new products by configuration	SLOs met (doc 03)

Why 8–10 weeks is credible. The business logic is already implemented and **rules-driven**: scoring, journeys, NBA, benefits, tariffs, contact policy and copy are configuration in `config/rules/*.json`. Every external system sits behind a port with a sandbox adapter (`src/adapters/integrations/mockGateways.js`, `simulatedCaller.js`). MVP effort therefore goes to integration, data onboarding, UAT and the pilot, not to building logic.

11. KPI tree, baselines and targets

11.1 KPI tree



11.2 KPI definitions

Targets are **proposed assumptions** for the pilot and scale phases. They must be re-baselined after Phase 0. "TBM" means to be measured in discovery.

KPI	Definition	Source in platform	Baseline (brief / TBM)	Pilot target (proposed)	Scale target Y1 (proposed)
K-00	Policies issued via own channels + partner API	<code>GET /api/dashboard/overview → sales.orders, byChannel</code>	Own-channel renewals = 0 this year (brief)	≥ 2,000 in pilot cohort	≈ 140,000 (base case §12)
K-01	% profiles with expiry confidence ≥ 0.5	<code>base.profilesWithUsableData</code> (see <i>engineering note E-12, doc 05</i>)	~ 10% valid stamp (brief)	35%	50%
K-02	% profiles reachable by push or Zalo	<code>profiles.channels</code>	TBM (70–80% app users per brief)	60%	65%
K-03	Expiry confirmations per 1,000 profiles per month	Audit <code>customer.expiry_declared</code>	0 (no capture today)	30	50
K-04	SEG-3 renewal rate	Orders with journey <code>renewal</code> / TASCO-book expiries	TBM	15%	25%
K-05	SEG-4 conversion	Orders with journey <code>conquest</code> / conquest audience	TBM	1%	2%
K-06	SEG-1 recovery	Orders with journey <code>lapsed_uninsured</code> / lapsed audience	TBM	3%	6%
K-07	SEG-2 conversion	Orders with journey <code>new_vehicle</code> / new tags	TBM	2%	4%
K-08	Bot hot-handoff rate; handoff win rate	<code>voice_sessions.outcome, handoffs.status</code>	Telesales conversion "poor" (brief)	8% / 20%	10% / 25%
K-09	PA_SEAT / MOTOR_PD attach on own-channel TNDs	<code>activePoliciesByProduct</code>	TBM	4% / 0.25%	8% / 0.5%
K-10	Variable acquisition cost per policy	<code>economics.costPerOrder + telesales</code>	TBM	≤ VND 55,000	≤ VND 40,000
K-11	Bot cost ÷ telesales-equivalent cost	<code>economics.voiceBotCost, equivalentTelesalesCost</code>	n/a	≤ 10%	≤ 10% (by unit costs: 8.9%)
K-12	Bot opt-out rate; plate-verification failure rate	<code>GET /api/dashboard/governance</code>	n/a	≤ 5%; ≤ 10%	≤ 4%; ≤ 8%
K-13	Copy-guard blocks in production; complaints per 10,000 contacts	<code>messageStatus.blocked</code> ; complaints feed (planned)	n/a	0 blocks after go-live; TBM	0; TBM
K-14	Audit chain verifies	<code>GET /api/audit/verify → ok: true</code>	n/a	100% daily	100% daily

Measurement design. The pilot should hold out a randomised **control group** (for example 10% of the cohort receiving no own-channel contact). Only then can incremental policies be separated from policies that partners

would have sold anyway.

12. Business case model (illustrative)

All numbers below are illustrative. Unit costs come from `config/rules/costs.json`, regulated premiums from `config/rules/tariff.tnds_car.json`, and illustrative PA and PD rates from `rating.pa_seat.json` and `rating.motor_pd.json`. Every other input is an assumption [A-*nn*]. FX assumption: **USD 1 = VND 26,000** [A-01].

12.1 Formulas

Actionable vehicles	$A = B \times e \times u$	
B = VETC car base		(≈ 6,000,000, brief)
e = share with an expiry event in the year		[A-02]
u = usable-expiry rate after data repair		[A-03]
Segment volumes	$A_{ren} = A \times s_T$; $A_{conq} = A \times s_C$; $A_{lap} = A \times s_L$	[A-04]
New vehicles per year	N	[A-05]
Own-channel TNDs policies	$Q_{own} = A_{ren} \cdot c_{ren} + A_{conq} \cdot c_{conq} + A_{lap} \cdot c_{lap} + N \cdot c_{new}$	[A-06]
Partner-API policies	Q_p	[A-07]
TNDs net premium	$GWP_{TNDs} = (Q_{own} + Q_p) \times \bar{P}$	[A-08]
Cross-sell premium	$GWP_X = Q_{own} \times (a_{PA} \times P_{PA} + a_{PD} \times P_{PD})$	[A-09]
Messaging cost	$C_{msg} = (A + N) \times m \times (0.3 \times 300 + 0.1 \times 700)$ (60% push @ 0)	[A-10]
Bot cost	$C_{bot} = A \times k \times 1.3 \times 1,500 \times 1.6$	[A-11]
Telesales cost	$C_{tel} = calls_{tel} \times 6,000 \times 4.5$	[A-12]
Partner commission (commission.json)	$C_{com} = Q_p \times \bar{P} \times 5\%$	
Acquisition cost per policy	$= (C_{msg} + C_{bot} + C_{tel} + C_{com}) / (Q_{own} + Q_p)$	

Premium inputs.

- $\bar{P}$ (average TNDs net premium, mix [A-08] : 70% car < 6 seats, 25% car 6–11 seats, 5% at VND 1,270,000) = $0.70 \times 437,000 + 0.25 \times 794,000 + 0.05 \times 1,270,000 = \text{VND } 567,900$ (VND 624,690 incl. VAT).
- $P_{PA} = 5 \text{ seats} \times \text{VND } 20,000,000 \times 0.1\% = \text{VND } 100,000$ (VAT-exempt per `rating.pa_seat.json`).
- $P_{PD} = \text{VND } 700,000,000 \text{ sum insured} \times 1.5\% (\text{age } 0\text{--}3) \times (1 - 10\% \text{ deductible relief}) = \text{VND } 9,450,000 \text{ net}$ (`rating.motor_pd.json`, which is labelled ILLUSTRATIVE).

12.2 Scenario inputs

Input	Conservative	Base	Stretch
e: share with expiry event [A-02]	85%	85%	85%
u: usable-expiry rate [A-03]	35%	50%	65%
Segment split $s_T / s_C / s_L$ [A-04]	10 / 85 / 5%	10 / 85 / 5%	10 / 85 / 5%
N: new tags per year [A-05]	250,000	250,000	250,000
$c_{ren} / c_{conq} / c_{lap} / c_{new}$ [A-06]	15% / 1% / 3% / 2%	25% / 2% / 6% / 4%	35% / 3% / 10% / 6%
Q_p : partner-API policies [A-07]	5,000	15,000	30,000
a_{PA} / a_{PD} attach [A-09]	4% / 0.25%	8% / 0.5%	12% / 1%

Input	Conservative	Base	Stretch
m: sends per profile per year [A-10]	5	5	5
k: share of A called by bot [A-11]	10%	15%	20%
Telesales calls per year [A-12]	25,000	50,000	80,000

12.3 Outputs (year 1 at scale)

Output	Conservative	Base	Stretch
Actionable vehicles A	1,785,000	2,550,000	3,315,000
SEG-3 renewals	26,775	63,750	116,025
SEG-4 conquest	15,172	43,350	84,532
SEG-1 lapsed recovered	2,678	7,650	16,575
SEG-2 new vehicles	5,000	10,000	15,000
SEG-7 partner API	5,000	15,000	30,000
Total TNDS policies	54,625	139,750	262,132
TNDS net premium (VND)	31.0 bn	79.4 bn	148.9 bn
TNDS net premium (USD)	1.19 M	3.05 M	5.73 M
PA_SEAT policies / premium (VND)	1,985 / 0.20 bn	9,980 / 1.00 bn	27,856 / 2.79 bn
MOTOR_PD policies / premium (VND)	124 / 1.17 bn	624 / 5.89 bn	2,321 / 21.94 bn
Total premium incl. cross-sell (VND)	32.4 bn	86.3 bn	173.6 bn
Messaging cost (VND)	1.63 bn	2.24 bn	2.85 bn
Bot calls / cost (VND)	232,050 / 0.56 bn	497,250 / 1.19 bn	861,900 / 2.07 bn
Same calls by telesales (VND)	6.27 bn	13.43 bn	23.27 bn
Telesales cost (VND)	0.68 bn	1.35 bn	2.16 bn
Partner commission (VND)	0.14 bn	0.43 bn	0.85 bn
Variable acquisition cost (VND)	3.00 bn	5.21 bn	7.93 bn
Per policy (VND)	54,955	37,276	30,261
As % of TNDS net premium	9.7%	6.6%	5.3%

12.4 Reading the numbers honestly

- The voice bot is the clearest saving.** At contracted unit costs, a bot call costs 8.9% of a telesales call (VND 2,400 vs VND 27,000). In the base case that is about **VND 12.2 bn per year** of qualifying work not done by humans.
- Own-channel acquisition cost per policy (6.6% of premium in the base case) is above the 5% TNDS commission benchmark.** Own channels do not win on cost per TNDS policy in year 1. They win on:
 - (a) **incremental volume** that partners do not reach (lapsed, in-app moments);
 - (b) **data ownership**, since renewals in year 2 need less verification and more free push;
 - (c) **cross-sell premium**, where PD alone adds about 7% to base-case premium;

- (d) **retention of the TASCO book**, currently zero through own channels. Target K-10 tracks the path toward parity.
3. **Incrementality must be measured.** Some own-channel policies would have been sold by partners anyway. The pilot control group (§11.2) quantifies this.
 4. **Platform cost** (doc 07: about USD 0.79 M in year 1, then about USD 0.50 M per year) must be covered by the **technical contribution margin** of incremental premium. TASCO finance should plug its own margin into: $\text{Break-even margin} = (\text{Platform cost} + \text{Acquisition cost}) / (\text{Incremental premium})$. In the base case year 2, assuming 100% incrementality, this is $(\text{VND } 13.1 \text{ bn} + 5.2 \text{ bn}) / 86.3 \text{ bn} \approx \mathbf{21\%}$. At 70% incrementality [A-13] it rises to about **30%**. This is why the pilot must prove incrementality before scale.

12.5 Sensitivity (base case, TND\$ net premium)

Driver changed (±)	Impact on TND\$ net premium
u (usable expiry) ±10 pts	±VND 13.0 bn (every segment conversion scales with A)
c_ren ±5 pts	±VND 7.2 bn
c_conq ±1 pt	±VND 12.3 bn
Q_p ±10,000	±VND 5.7 bn

The **data-repair rate (u)** and **conquest conversion** dominate. That is why the strategy puts data repair (fix_data NBA) ahead of selling.

13. Assumptions register

ID	Assumption	Value used	Owner to validate
A-01	FX rate	USD 1 = VND 26,000	TASCO finance
A-02	Share of base with a TND\$ expiry event in the year (rest on multi-year or unknown)	85%	Discovery (data)
A-03	Usable-expiry rate achievable after data repair	35% / 50% / 65%	Pilot
A-04	Split of actionable base: TASCO book / elsewhere or unknown / lapsed	10% / 85% / 5%	Discovery (data)
A-05	New ETC tag activations per year (car)	250,000	VETC
A-06	Segment conversion rates	See §12.2	Pilot (control group)
A-07	Partner-API policies in year 1	5k / 15k / 30k	Partner managers
A-08	TND\$ category mix	70% / 25% / 5%	Discovery (data)
A-09	Cross-sell attach and average PD sum insured (VND 700 M)	See §12.2	TASCO product / actuarial
A-10	Sends per profile per year; channel mix 60% push / 30% ZNS / 10% SMS	5	Campaign manager

ID	Assumption	Value used	Owner to validate
A-11	Share of actionable called by bot; 1.3 attempts per call	10–20%	Campaign manager
A-12	Telesales calls per year	25k / 50k / 80k	Telesales lead
A-13	Incrementality vs partner channel	70% (sensitivity)	Pilot control group
A-14	Tariff, commission caps and contact policy as encoded in rules	As in config/rules	TASCO underwriting, finance, legal

14. Fit with the judging criteria

Criterion	How the proposal scores	Evidence
Impact	Attacks all four root causes (data, trust, value, friction) across seven segments , not only renewal. Base case about 140k policies and VND 86 bn premium.	\$4, \$12; <code>journeys.json</code> (4 journeys + cross-sell); partner API
Fit	Built on Tasco group assets (VETC app reach, wallet, tag events, inspection, TASCO issuance). Respects regulated pricing (copy guard, no discount field in quotes) and partner economics (statutory caps). Supports VETC's multi-insurer model.	<code>copy_guard.json</code> , <code>rating.js</code> (<code>priceRegulated</code> note), <code>commission.json</code>
Speed	Working codebase with 73 API routes, rules-driven logic and sandbox ports. MVP pilot in 8–10 weeks. Rule changes in hours via maker-checker with no code release.	<code>src/adapters/http/routes.js</code> , <code>rulesService.js</code> ; adoption target "rule change lead time ≤ 1 day" (<code>insightsService.adoption</code>)
Cost	Free push first; bot at 8.9% of telesales cost per call; serverless-friendly Node.js and PostgreSQL stack; no licensed rules engine.	<code>costs.json</code> ; <code>package.json</code> (single runtime dependency pg)

15. What was added beyond the brief

#	Addition	Why it matters	Where
1	New-business journeys: lapsed and uninsured, new vehicle, conquest	Growth beyond the small TASCO book	<code>journeys.json</code>
2	Event-driven moments of truth (tag activation, inspection booking, wallet top-up, long trip)	Contact when the driver is receptive and in the app	<code>triggers.json</code> , POST <code>/api/ecosystem/events</code>
3	Partner API and commission statements	Arms the channel that already closes most deals	<code>/api/partner/v1/*</code>
4	Multi-product catalogue and bundles (MOTOR_PD, PA_SEAT, SAFE_DRIVE, FULL_MOTOR)	Raises value per customer without discounting	<code>products.json</code> , <code>rating.*.json</code>
5	Public QR certificate verification	Trust, and proof of cover at roadside checks	<code>/verify/:certNo</code>
6	Claims FNOL in the app	Trust is earned at claim time	POST <code>/api/customer/claims</code>

#	Addition	Why it matters	Where
7	Rules governance with maker-checker, simulation and rollback	The business changes scoring, journeys and copy safely without a release	<code>/api/rules/*</code>
8	Copy guard	Discount language cannot reach customers	<code>copy_guard.json</code> , <code>validators.js</code>
9	Contact policy engine (consent, DNC, hours, caps)	Anti-spam compliance by construction	<code>contact_policy.json</code> , <code>contactPolicy.js</code>
10	Data lineage, DQ queue, steward corrections	Data repair is operational, not ad hoc	<code>/api/dq/*</code> , <code>/api/customers/:id/lineage</code>
11	Tamper-evident audit trail (hash chain)	Regulator-grade accountability	<code>auditChain.js</code> , <code>GET /api/audit/verify</code>
12	Data-subject rights (export, erase or anonymise, consent centre)	PDP law readiness	<code>/api/dsar/*</code> , <code>/api/customer/data-export</code>
13	Economics dashboard using contracted unit costs	Cost per order and bot vs human saving visible daily	<code>GET /api/dashboard/overview → economics</code>
14	Multi-year cover aligned to the inspection cycle	Fewer renewals to win; convenience	<code>Benefit multi_year</code> ; <code>tariff maxTermYears: 3</code>

16. Key risks and mitigations

ID	Risk	Likelihood	Impact	Mitigation
R-01	VETC data-sharing or consent basis insufficient for marketing use	Medium	High	Phase 0 legal review; consent captured per channel; service messages separated from marketing (<code>marketing</code> flag per step)
R-02	Multi-insurer neutrality: VETC promotes TASCO over other insurers on its platform	Medium	High	Group governance decision; the platform can present the regulated price as "same everywhere"; benefits are service-based; legal review of copy
R-03	Partner conflict (perception that VETC competes with partners)	Medium	Medium	Partner API with instant issuance and transparent commission; own channels focus on moments partners do not see
R-04	Voice bot ASR accuracy on Vietnamese plates and dialects	Medium	Medium	Plate-first verification with 3 attempts; fallback to sending a link; vendor bake-off in Phase 0; governance KPIs (K-12)
R-05	Copy or regulatory change (tariff, commission caps)	Low	Medium	All in rules with maker-checker; validators enforce caps
R-06	Low incrementality vs partners	Medium	High	Control group; segment focus on lapsed and in-app moments
R-07	Loyalty or referral not approved by legal	Medium	Low	Already gated: <code>legalStatus: pending_legal_review</code> ; <code>referral.enabled: false</code>

ID	Risk	Likelihood	Impact	Mitigation
R-08	Company-owned vehicles also enter B2C journeys (journey audiences do not exclude ownerType = company ; only the NBA routes them to B2B)	High (as built)	Medium	Engineering fix proposed (doc 05, E-07)
R-09	Data residency requirements for personal data	Medium	High	Deploy in a Vietnam region or on Vietnamese cloud if required; confirm with TASCO legal (doc 03 NFR-030)